Step 2b - H3 - Microwave user efficiency simulation

Tech baseline

Technologies used in the market

Multiphysics EM-thermal co-simulation (COMSOL RF+Heat Transfer, Ansys HFSS+thermal, CST Studio)

How it works: Full-wave frequency- or time-domain EM solver coupled to heat transfer, iterated as dielectric properties change with temperature; used offline to test what-if scenarios (load geometry, power level, belt position).

How well it performs: Experimentally validated for applicators (e.g., 915 MHz waveguide vs thermocouple data); single cavity runs take minutes-hours, full parametric sweeps days; expert setup effort measured in engineer-weeks

Limits: Requires scarce EM expertise + expensive seats ($10-70k); accuracy collapses without temperature-dependent dielectric data for the actual product; does not connect to machine telemetry — it is a study tool, not an operations tool

Dedicated microwave-heating FDTD simulation (QWED QuickWave + Basic Heating Module)

How it works: FDTD co-simulation with temperature-/enthalpy-dependent media, moving/rotating loads, source-frequency tuning and source switching — the only commercial solver family purpose-built for microwave heating.

How well it performs: Handles realistic process features (conveyor motion, load rotation, thermal runaway); runs on workstations; smaller vendor so limited global support footprint

Limits: Still an engineering tool needing trained users and per-material dielectric input; no built-in library of operator-facing KPIs or machine templates; minimal presence outside specialist circles

Generic process digital twins / APC platforms (AVEVA, AspenTech, Siemens, Yokogawa)

How it works: Plant-data-driven models, energy dashboards and advanced process control applied to thermal lines; the microwave unit treated as a black-box actuator.

How well it performs: Process digital-twin market ~$2.6B (2025) growing ~31%/yr; proven for furnaces/dryers; delivers ~5-15% typical energy savings in conventional thermal processes

Limits: Cannot model EM field distribution, hot spots or dielectric coupling — the very phenomena that determine microwave efficiency; treat the microwave stage as unmodelled equipment

Solid-state microwave generators with closed-loop control (RFHIC GaN, LEANFA/RFMW, SAIREM, Muegge)

How it works: Replace/retrofit magnetrons with LDMOS/GaN amplifiers enabling real-time control of power, frequency and phase, often fed by reflected-power sensing.

How well it performs: Solid-state systems claim >70% efficiency vs ~45-50% for magnetrons and >15-yr device life vs ~5,000-h magnetron warranty; enable automatic power matching under load variation

Limits: Hardware capex, not software; benefits materialize only when combined with an operating model telling it what to do; retrofit business case still developing

Incumbent baseline: operator heuristics + OEM service contracts + trial-and-error

How it works: Operators tune power/belt speed/recipe empirically; OEMs provide occasional optimization visits; universities/consultants run one-off studies.

How well it performs: Zero software cost but leaves 10-30% energy/quality upside unrealized; microwave drying studies report up to ~30% energy savings and up to ~80% shorter drying times vs conventional tunnels when properly optimized

Limits: Knowledge leaves with retiring specialists; empiricism cannot see inside the cavity (hot spots, arcing risk, runaway); inconsistent across batches/materials

Why our solution can be better

Incumbents fail this hypothesis in two directions: generalist EM tools can compute the physics but require experts the operators do not employ, and generic digital-twin platforms reach the operator but are blind to microwave physics. A domain-focused tool built on ProSeed's microwave know-how — pre-modelled machine/applicator templates, a validated dielectric database, and operator-facing KPIs (energy per kg product, uniformity index, runaway margin) — can deliver simulation-driven setpoint optimization without requiring an RF engineer on staff. ProSeed can also pair the software with retrofit sensing (reflected-power, temperature) so models self-calibrate on the real machine, something neither simulation vendors nor twin platforms currently do.

Key assumptions

• Global installed base of industrial microwave process machines is ~10^4-10^5 units (assumed; anchored to ~$1.4-1.6B/yr equipment sales and SAIREM's disclosed ~5,000 installed machines alone)

• Industrial microwaves in Europe consume ~9.6 TWh/yr and typical magnetron systems run at ~45-50% wall-to-load efficiency vs >70% claimed for solid-state (measured, EU H2020 CLEAN-HEAT)

• Realistic efficiency/quality gain from simulation-guided operation is ~10-30% energy and less scrap/hot-spot damage (assumed; supported by reported ~30% tunnel-dryer savings and dielectric-sensitivity literature)

• Operators will not hire RF specialists: product must be usable by process engineers, which caps acceptable complexity (assumed)

• Per-site value requires measured dielectric/thermal data of the actual product — a service-heavy calibration step that limits pure-SaaS scalability (assumed, consistent with literature emphasis on temperature-dependent dielectric data scarcity)

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Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis